

HMS 581: FINAL REPORT

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1. INTRODUCTION AND BACKGROUND

Measles is a highly contagious disease which spreads through droplets from coughing, sneezing, or close contact with an infected individual. Characterized by symptoms such as fever, cough, runny nose, and a distinctive rash, measles can lead to severe complications such as pneumonia or encephalitis. In extreme cases, measles can even cause death particularly in young children and immunocompromised individuals [1].

The basic reproduction number r_0 of measles is among the highest of any known infectious disease, typically estimated between 12 and 18 [4]. This means that in a fully susceptible population, a single case can generate up to 18 secondary infections. The high r_0 reflects the extreme transmission rate of the virus and explains the need for a high coverage of vaccination to achieve herd immunity.

In mathematical models, the transmission rate β is related to r_0 via the formula:

$$\beta = r_0 \cdot \gamma,$$

where γ is the recovery rate. Before the introduction of the measles vaccine in 1963, the disease was widespread in the United States, with many children exposed to it. Its predictable seasonal epidemics make it a great case study for mathematical modeling of infectious diseases.

2. MODEL DESCRIPTION

The model was built by using reported case data from California and Nevada. It focused specifically on the pre-vaccine era from 1945 to 1960, during which measles was endemic and followed a seasonal pattern. The original data included weekly case counts which were aggregated to monthly totals to match the model's discrete time step of one month. In a few instances, missing data entries occurred during lull periods between outbreaks. These gaps were filled using the average of the neighboring months. This typically resulted in a low case estimate for that period, but helped preserved continuity without inflating transmission (Figure 1).

To improve the biological realism and fit of the model, three augmentations were introduced to the SIR framework developed within class:

- **Seasonality of Transmission:** The transmission rate β was controlled by a sinusoidal seasonal forcing term to reflect the winter and spring peaks in measles cases.
- **Birth and Death Rates:** Constant birth and death rates were included to allow for demographic turnover in the susceptible population and realistic long-term dynamics as populations numbers exploded in the given time period.
- **State-to-State Migration:** Individuals were allowed to migrate between CA and NV compartments (S, I, and R) at fixed rates in an attempt to account for coupling of the epidemics in each state.

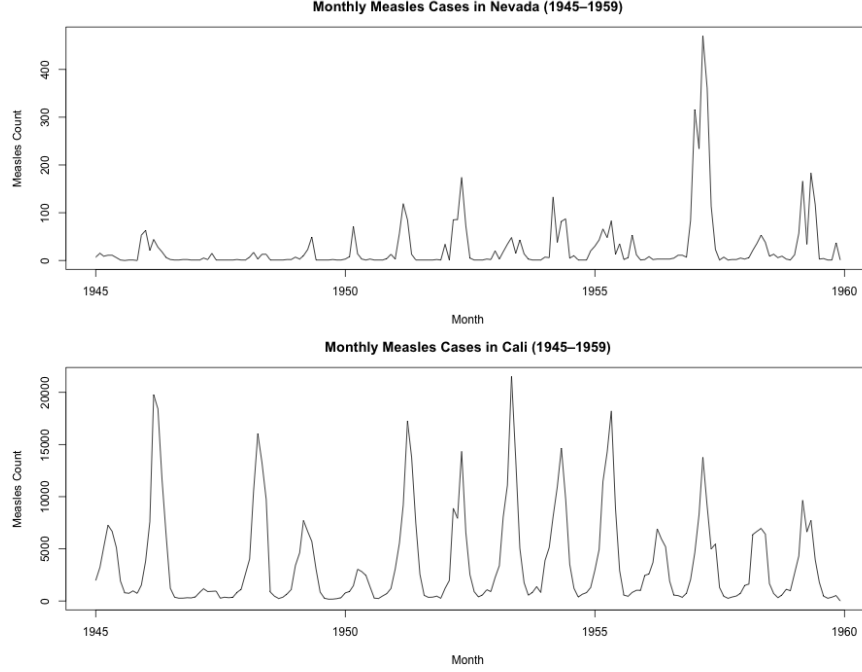


FIGURE 1. Monthly Measles Cases in CA and NV during specified time period.

Together, these augmentations enabled the model to better capture the observed patterns in the case data and represent more realistic transmission dynamics across time and space.

3. RESULTS AND PARAMETER ESTIMATION

To estimate the parameters of the two-region stochastic SIR model, a Metropolis-Hastings Markov Chain Monte Carlo (MCMC) algorithm was implemented. This involved drawing proposals from a normal distribution and accepting or rejecting these proposals based on their likelihood under the model. This included constructing a target distribution based on observed case counts and defining a reasonable proposal distribution for each parameter. Before running the MCMC algorithm, initial guesses were generated and evaluated for the parameters based on the known biological and demographic properties of measles and the populations of California and Nevada:

- **Transmission and Recovery Rates (β, γ):** The relationship $r_0 = \beta/\gamma$ was used to estimate initial values. With $r_0 \approx 15$ and an known infectious period of about 2 weeks ($\gamma \approx 52/2 = 26$ per year), this gave an initial estimate of

$$\beta = r_0 \cdot \gamma \approx 15 \cdot 26 = 390.$$

A random number was selected as an initial guess around this range. Additionally, each state was given a respective beta. Although beta should be constant across both states, multiple runs showed that better fits were produced for each state having a unique β that was close to the theoretical calculated one for fitting purposes.

- **Seasonality Parameters:** Measles shows strong seasonal trends, so a seasonal forcing amplitude (sea) and phase shift (peak) were included. Based on the data, peak transmission typically occurred around March, corresponding to a phase of about 3 months. We chose an initial amplitude allowing for strong but not overwhelming seasonal variation so the model would not die out.

- Initial Conditions (S_0, I_0): We assumed populations were near endemic equilibrium due to measles' high r_0 , giving a recovered population R_0 estimate:

$$R_0 \approx 1 - \frac{1}{r_0} \approx 1 - \frac{1}{15} = 0.933.$$

The susceptible pool would almost make up the remaining percentage of the population from this pool. The infectious fraction was assumed to be small but nonzero, reflecting ongoing transmission in the population.

- Reporting Rate (ρ): Since not all cases were observed, the reporting rate ρ was allowed to vary in the range $(0, 1)$ and estimated it from data.
- Migration Rates: Migration between states was assumed to be low, so initial guesses for migration parameters were small to prevent rapid population mixing.
- Birth and Death Rates (μ, δ): These were estimated from historical population growth data from the Federal Reserve Bank of Saint Louis [2] & [3]. Given population estimates from 1945 to 1960, (r), the monthly birth rates were estimated using:

$$N(t) = N_0 e^{\frac{r}{12}t}$$

State specific population values were used to calculate and then a reasonable range around these rates were applied for both birth and death.

Initial parameter vectors were generated until one yielding a non-negative, non-NaN value for the target distribution was found to start the MCMC algorithm. Chains were then run to sample from the posterior distribution of parameters given the observed measles case data in both California and Nevada.

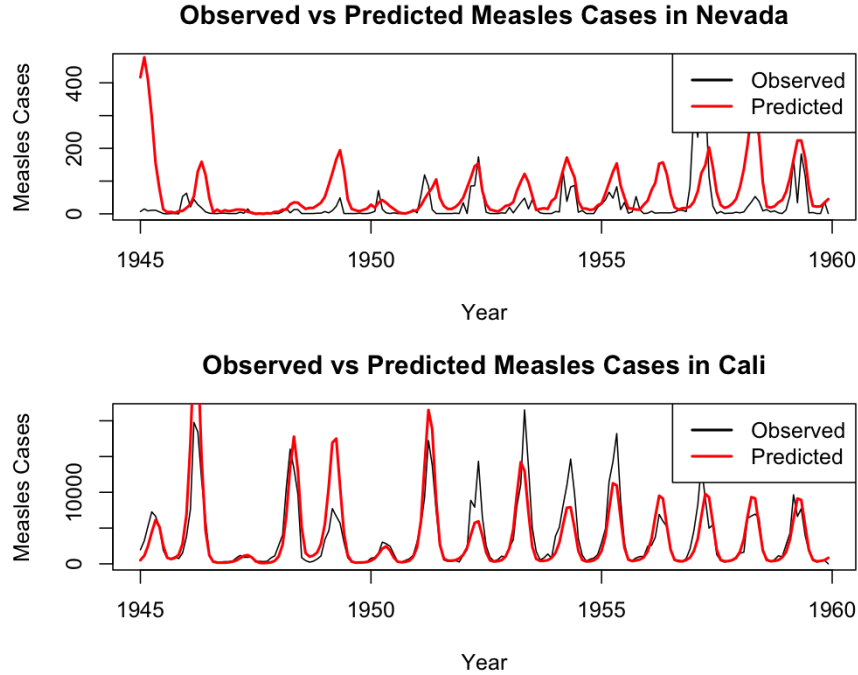


FIGURE 2. Predicted versus Observed Cases for Each State.

4. DISCUSSION

One of the key modeling choices that emerged during fitting was the decision to allow separate transmission rates (β) for California and Nevada. While it initially seemed ideal to assume a single

β value for both states, this produced consistently poor fits for one of the states. Allowing distinct transmission rates improved the fit for both states' observed case curves, but there is not a strong biological rationale for this decision outside of optimization.

Parameter fitting proved challenging. Despite numerous iterations of the MCMC procedure, this final parameter set represents the best-performing configuration obtained. However, this came at the cost of reducing the effective coupling between the states through state transmission. The current model's structure and parameter space made it difficult to simultaneously tune both local transmission dynamics and cross-state infection flow without sacrificing overall model stability.

A valuable extension to the model would be the inclusion of age-structured compartments. Age is a known critical factor in measles transmission dynamics due to age-specific susceptibility fractions which could include contact patterns. This would be important particularly in pre-vaccine eras where childhood infections dominated. This would also create opportunities to explore the impact of targeted interventions and better explain variations in observed case counts.

REFERENCES

- [1] Shinta R Arsal, Dipo Aldila, and Bevina D Handari. Short review of mathematical model of measles. In *AIP Conference Proceedings*, volume 2264. AIP Publishing, 2020.
- [2] Federal Reserve Bank of St. Louis. Resident Population in California, 2024. Accessed: 2025-06-08.
- [3] Federal Reserve Bank of St. Louis. Resident Population in Nevada, 2024. Accessed: 2025-06-08.
- [4] Fiona M Guerra, Shelly Bolotin, Gillian Lim, Jane Heffernan, Shelley L Deeks, Ye Li, and Natasha S Crowcroft. The basic reproduction number (r_0) of measles: a systematic review. *The Lancet Infectious Diseases*, 17(12):e420–e428, 2017.

APPENDIX A. FIGURE STATISTICS

Parameter	Description	Value
β_{CA}	Transmission rate (California)	577.13
β_{NV}	Transmission rate (Nevada)	338.15
γ	Recovery rate	14.25
a	Seasonality amplitude	0.831
ϕ	Seasonal peak phase	0.067
S_0^{CA}	Initial susceptible fraction (CA)	0.020
I_0^{CA}	Initial infectious fraction (CA)	8.53e-05
ρ_{CA}	Reporting rate (CA)	0.404
$M_{CA \rightarrow NV}$	Migration rate CA $\rightarrow$ NV	0.0035
μ_{CA}	Birth rate (CA)	0.0150
δ_{CA}	Death rate (CA)	0.0074
S_0^{NV}	Initial susceptible fraction (NV)	0.0272
I_0^{NV}	Initial infectious fraction (NV)	0.0064
ρ_{NV}	Reporting rate (NV)	0.406
$M_{NV \rightarrow CA}$	Migration rate NV $\rightarrow$ CA	0.0027
μ_{NV}	Birth rate (NV)	0.0019
δ_{NV}	Death rate (NV)	0.0014

TABLE 1. Posterior parameter estimates from the MCMC in natural space for California and Nevada Measles data (1945–1960).

Best Log-Likelihood: -84067.38

APPENDIX B. CODE

```

1
2 rm(list = ls())
3 set.seed(1)
4
5
6 require(glue)
7 require(MASS)
8 require(mvtnorm)
9 require(faraway)
10
11 setwd("/Users/jaxontuggle/Downloads")
12 measles <- read.csv("measles_Jaxson.csv")[, -1]
13
14 measles$StartDate <- as.Date(measles$PeriodStartDate, "%m/%d/%Y")
15 measles$EndDate <- as.Date(measles$PeriodEndDate, "%m/%d/%Y")
16 measles$row_duration <- as.numeric(measles$EndDate - measles$StartDate)
17
18 #Organization into Nevada and Cali data subsets monthly from 1945-1960
19
20 nevsubset <- measles[which(measles$StartDate >= as.Date("1945-01-01") &
21                           measles$EndDate <= as.Date("1959-12-01") &
22                           measles$Admin1Name == "NEVADA"),]
23 calsubset <- measles[which(measles$StartDate >= as.Date("1945-01-01") &
24                           measles$EndDate <= as.Date("1959-12-01") &
25                           measles$Admin1Name == "CALIFORNIA"),]
26
27 all_months <- seq(as.Date("1945-01-01"), as.Date("1959-12-01"), by = "
    month")
28 all_months_fmt <- format(all_months, "%Y-%m")
29 full_months_df <- data.frame(YearMonth = all_months_fmt)
30
31 nevsubset$YearMonth <- format(nevsubset$StartDate, "%Y-%m")
32 nev_data <- aggregate(CountValue ~ YearMonth, data = nevsubset, sum)
33 nev_data_full <- merge(full_months_df, nev_data, by = "YearMonth", all.x =
    TRUE)
34 nev_data_full$CountValue[is.na(nev_data_full$CountValue)] <- 1
35 NV_obs_data <- nev_data_full$CountValue
36
37
38 calsubset$YearMonth <- format(calsubset$StartDate, "%Y-%m")
39 cal_data <- aggregate(CountValue ~ YearMonth, data = calsubset, sum)
40 cal_data_full <- merge(full_months_df, cal_data, by = "YearMonth", all.x =
    TRUE)
41 cal_data_full$CountValue[is.na(cal_data_full$CountValue)] <- 0
42 CA_obs_data <- cal_data_full$CountValue
43
44
45 # Plots of each data
46
47 par(mfrow = c(2, 1), mar = c(4, 4, 3, 1))
48 nev_data_full$MonthDate <- as.Date(paste0(nev_data_full$YearMonth, "-01"))
49 plot(nev_data_full$MonthDate, nev_data_full$CountValue, type = 'l',

```

```

50     xlab = "Month", ylab = "Measles_Count",
51     main = "Monthly_Measles_Cases_in_Nevada_(1945 1959 )")
52
53 cal_data_full$MonthDate <- as.Date(paste0(cal_data_full$YearMonth, "-01"))
54 plot(cal_data_full$MonthDate, cal_data_full$CountValue, type = 'l',
55       xlab = "Month", ylab = "Measles_Count",
56       main = "Monthly_Measles_Cases_in_Cali_(1945 1959 )")
57
58
59 #Constants
60 # Gotten from information FRED
61 Population_CA <- 9.344e6
62 Population_NV <- 149000
63
64 NumYears <- 15
65 Year_start <- 1945
66 Year_end <- Year_start + NumYears - 1
67 Date_start <- as.Date(glue("{Year_start}-01-01"))
68 Date_end <- as.Date(glue("{Year_end}-12-31"))
69 time_step = "1_month"
70 Dates <- seq(Date_start, Date_end, by = time_step)
71 dt <- ifelse(time_step == "1_year", 1, ifelse(time_step == "1_month", 1/
       12, ifelse(time_step == "1_week", 1/52, 1/365)))
72
73 # Needed Functions
74 logit_bounds <- function(a, b) {
75   logit_a <- log(a / (1 - a))
76   logit_b <- log(b / (1 - b))
77   runif(1, logit_a, logit_b)
78 }
79
80 ilogit <- function(x) 1 / (1 + exp(-x))
81
82
83 beta <- function(date, par){
84   return(par)
85 }
86
87 transform_pars <- function(pars){
88   out <- c(exp(pars[1]),
89           exp(pars[2]),
90           exp(pars[3]),
91           #CA
92           ilogit(pars[4]),
93           ilogit(pars[5]),
94           ilogit(pars[6]),
95           ilogit(pars[7]),
96           ilogit(pars[8]),
97           ilogit(pars[9]),
98           exp(pars[10]),
99           exp(pars[11]),
100          #NV
101          ilogit(pars[12]),
102          ilogit(pars[13]),

```

```

103         ilogit(pars[14]),
104         ilogit(pars[15]),
105         exp(pars[16]),
106         exp(pars[17]))
107     return(out)
108 }
109
110 SIR_TWO_STATE_MODEL <- function(pars) {
111     # Transform
112     pars_t <- transform_pars(pars)
113
114     # Ind Pars
115     beta_CA <- pars_t[1]
116     beta_NV <- pars_t[2]
117     gamma <- pars_t[3]
118     sea <- pars_t[4]
119     pea <- 12 * pars_t[5]
120
121     # CA
122     S0_frac_CA <- pars_t[6]
123     I0_frac_CA <- pars_t[7]*(1 - S0_frac_CA)
124     rho_CA <- pars_t[8]
125     mig_CA_to_NV <- pars_t[9]
126     mu_CA <- pars_t[10]
127     delta_CA <- pars_t[11]
128
129     # NV
130     S0_frac_NV <- pars_t[12]
131     I0_frac_NV <- pars_t[13] * (1 - S0_frac_NV)
132     rho_NV <- pars_t[14]
133     mig_NV_to_CA <- pars_t[15]
134     mu_NV <- pars_t[16]
135     delta_NV <- pars_t[17]
136
137     # ICs
138     S_CA <- round(S0_frac_CA * Population_CA)
139     I_CA <- round(I0_frac_CA * Population_CA)
140     R_CA <- Population_CA - S_CA - I_CA
141
142     S_NV <- round(S0_frac_NV * Population_NV)
143     I_NV <- round(I0_frac_NV * Population_NV)
144     R_NV <- Population_NV - S_NV - I_NV
145
146     out <- array(NA, dim = c(length(Dates) + 1, 2, 7))
147     out[1, 1, ] <- c(S_CA, I_CA, R_CA, 0, 0, 0, NA)
148     out[1, 2, ] <- c(S_NV, I_NV, R_NV, 0, 0, 0, NA)
149
150     # --- Time loop ---
151     for (d_num in seq_along(Dates)) {
152         # Vals by State
153         S_CA <- out[d_num, 1, 1]; I_CA <- out[d_num, 1, 2]; R_CA <- out[d_num,
154             1, 3]
155         S_NV <- out[d_num, 2, 1]; I_NV <- out[d_num, 2, 2]; R_NV <- out[d_num,
156             2, 3]

```

```

155 N_CA <- S_CA + I_CA + R_CA
156 N_NV <- S_NV + I_NV + R_NV
157
158 # Beta
159 beta_t_CA <- beta_CA * (1 + sea * sin(2 * pi * (d_num + pea) / 12))
160 beta_t_NV <- beta_NV * (1 + sea * sin(2 * pi * (d_num + pea) / 12))
161
162 #Probs
163 p_inf_CA <- 1 - exp(-beta_t_CA * I_CA / N_CA * dt)
164 p_rec_CA <- 1 - exp(-gamma * dt)
165 p_inf_NV <- 1 - exp(-beta_t_NV * I_NV / N_NV * dt)
166 p_rec_NV <- 1 - exp(-gamma * dt)
167
168 #Infs
169 new_inf_CA <- rbinom(1, S_CA, p_inf_CA)
170 new_rec_CA <- rbinom(1, I_CA, p_rec_CA)
171 new_case_CA <- rbinom(1, new_inf_CA, rho_CA)
172
173 new_inf_NV <- rbinom(1, S_NV, p_inf_NV)
174 new_rec_NV <- rbinom(1, I_NV, p_rec_NV)
175 new_case_NV <- rbinom(1, new_inf_NV, rho_NV)
176
177 # B/D
178 births_CA <- rpois(1, mu_CA * N_CA * dt)
179 births_NV <- rpois(1, mu_NV * N_NV * dt)
180
181 deaths_S_CA <- rbinom(1, S_CA, delta_CA * dt)
182 deaths_I_CA <- rbinom(1, I_CA, delta_CA * dt)
183 deaths_R_CA <- rbinom(1, R_CA, delta_CA * dt)
184 deaths_S_NV <- rbinom(1, S_NV, delta_NV * dt)
185 deaths_I_NV <- rbinom(1, I_NV, delta_NV * dt)
186 deaths_R_NV <- rbinom(1, R_NV, delta_NV * dt)
187
188 #Migration
189 mig_S_CA_to_NV <- rbinom(1, S_CA, mig_CA_to_NV)
190 mig_I_CA_to_NV <- rbinom(1, I_CA, mig_CA_to_NV)
191 mig_R_CA_to_NV <- rbinom(1, R_CA, mig_CA_to_NV)
192 mig_S_NV_to_CA <- rbinom(1, S_NV, mig_NV_to_CA)
193 mig_I_NV_to_CA <- rbinom(1, I_NV, mig_NV_to_CA)
194 mig_R_NV_to_CA <- rbinom(1, R_NV, mig_NV_to_CA)
195
196 # Updates
197 S_CA_next <- S_CA - new_inf_CA + births_CA - deaths_S_CA -
198   mig_S_CA_to_NV + mig_S_NV_to_CA
199 I_CA_next <- I_CA + new_inf_CA - new_rec_CA - deaths_I_CA -
200   mig_I_CA_to_NV + mig_I_NV_to_CA
201 R_CA_next <- R_CA + new_rec_CA - deaths_R_CA -
202   mig_R_CA_to_NV + mig_R_NV_to_CA
203
204 S_NV_next <- S_NV - new_inf_NV + births_NV - deaths_S_NV -
205   mig_S_NV_to_CA + mig_S_CA_to_NV
206 I_NV_next <- I_NV + new_inf_NV - new_rec_NV - deaths_I_NV -
207   mig_I_NV_to_CA + mig_I_CA_to_NV
208 R_NV_next <- R_NV + new_rec_NV - deaths_R_NV -

```

```

209     mig_R_NV_to_CA + mig_R_CA_to_NV
210
211     out[d_num + 1, 1, ] <- c(S_CA_next, I_CA_next,
212                             R_CA_next, new_inf_CA,
213                             new_rec_CA, new_case_CA,
214                             beta_t_CA)
215     out[d_num + 1, 2, ] <- c(S_NV_next, I_NV_next,
216                             R_NV_next, new_inf_NV,
217                             new_rec_NV, new_case_NV,
218                             beta_t_NV)
219
220     out[d_num, 1, 4:6] <- c(new_inf_CA, new_rec_CA, new_case_CA)
221     out[d_num, 2, 4:6] <- c(new_inf_NV, new_rec_NV, new_case_NV)
222     out[d_num, 1, 7] <- beta_t_CA
223     out[d_num, 2, 7] <- beta_t_NV
224 }
225
226 out <- out[-(length(Dates) + 1), , ]
227 return(out)
228 }
229
230 #ATTEMPT AT MCMC FOR MY MODEL
231
232 # Proposal Function
233 proposal_distribution <- function(proposal_mean, proposal_sd) {
234   rnorm(length(proposal_mean), mean = proposal_mean, sd = proposal_sd)
235 }
236
237 target_distribution <- function(pars){
238   guess_out <- SIR_TWO_STATE_MODEL(pars)
239   pars_t <- transform_pars(pars)
240
241   # Predicted new cases
242   pred_cases_CA <- guess_out[,1,6]
243   pred_cases_NV <- guess_out[,2,6]
244
245   loglik_CA <- sum(dpois(CA_obs_data, lambda = pred_cases_CA, log = TRUE)
246                   )
247   loglik_NV <- sum(dpois(NV_obs_data, lambda = pred_cases_NV, log = TRUE))
248   return(loglik_CA + loglik_NV)
249 }
250
251 # MH Step
252 MH_step <- function(log_p_new, log_p_current){
253   log_alpha <- log_p_new - log_p_current
254   log_r <- log(runif(1))
255   return(log_r < log_alpha)
256 }
257
258 # MCMC Step
259 MCMC_step <- function(current_pars, log_p_current, proposal_sd){
260   new_pars <- proposal_distribution(current_pars, proposal_sd)
261   log_p_new <- target_distribution(new_pars)

```

```

262
263 move <- MH_step(log_p_new, log_p_current)
264
265 if (move) {
266   return(list(new_pars, log_p_new))
267 } else {
268   return(list(current_pars, log_p_current))
269 }
270 }
271
272 # MCMC Loop
273 n_pars <- 17
274 proposal_sd <- rep(0.1, n_pars)
275
276 ## Herd Immunity: We know HI  $1 - 1/r_0$  (measles  $r_0 \sim 15$ ) so around 93% of pop
    must be recovered in this model meaning S0 and I0 must both be below 7
    percent at least!!
277
278 for (i in 1:100) {
279   current_pars <- initial_guess <- c(
280     log(runif(1,300,500)),          # log(beta_CA)
281     log(runif(1,300,500)),          # log(beta_NV)
282     log(52 / 2),                    # log(gamma)
283
284     logit_bounds(0.6, .8),          # sea
285     logit_bounds(0.2, .4),          # peak
286
287     logit_bounds(0.01, 0.05),       # S0_frac_CA
288     logit_bounds(1e-6, 0.01),       # I0_frac_CA
289     logit_bounds(0.4, 0.7),         # rho_CA
290     logit_bounds(0.0001, 0.01),     # mig_CA_to_NV
291
292     # CA params on log scale (exp-ed)
293     log(runif(1,.002, .006)),       # mu_CA
294     log(runif(1,.001, .003)),       # delta_CA
295
296     # NV params on logit scale
297     logit_bounds(0.01, 0.05),       # S0_frac_NV
298     logit_bounds(1e-6, 0.01),       # I0_frac_NV
299     logit_bounds(0.4, 0.7),         # rho_NV
300     logit_bounds(0.0001, 0.01),     # mig_NV_to_CA
301
302     # NV params on log scale
303     log(runif(1,.002, .006)),       # mu_NV
304     log(runif(1,.001, .003)),       # delta_NV
305
306
307 loglik <- target_distribution(current_pars)
308
309 if (is.finite(loglik) && !is.nan(loglik)) {
310   valid_params_found <- TRUE
311   best_initial_pars <- current_pars
312   best_initial_loglik <- loglik
313   cat("Valid parameters found on attempt", i, "\n")

```

```

314     break
315   } else {
316     cat("Attempt", i, "failed.\n")
317   }
318 }
319 p_current <- best_initial_loglik
320 out<- SIR_TWO_STATE_MODEL(best_initial_pars)
321 plot(out[,1,6])
322
323 N_chain <- 1000
324 all_pars <- matrix(NA, nrow = N_chain, ncol = n_pars)
325 all_logp <- numeric(N_chain)
326
327 all_pars[1, ] <- current_pars <- initial_guess
328 all_logp[1] <- p_current
329
330 for (i in 2:N_chain) {
331   step <- MCMC_step(current_pars, p_current, proposal_sd)
332   current_pars <- step[[1]]
333   p_current <- step[[2]]
334
335   all_pars[i, ] <- current_pars
336   all_logp[i] <- p_current
337
338   if (i %% 100 == 0) cat("Step:", i, "Log-Likelihood:", round(p_current,
339     2), "\n")
340 }
341
342 #Take the last pars that were valid before hitting 0
343 mcmc_pars <- all_pars[1000,]
344 mcmc_fit <- all_logp[1000]
345 par <- transform_pars(mcmc_pars)
346 out <- SIR_TWO_STATE_MODEL(mcmc_pars)
347 cases_ca <- out[,1,6]
348 cases_nv <- out[,2,6]
349
350 #Graphs
351 # Plot observed Nevada measles counts
352 par(mfrow = c(2, 1), mar = c(4, 4, 3, 1))
353 plot(nev_data_full$MonthDate, nev_data_full$CountValue, type = 'l', col =
354   "black",
355   xlab = "Year", ylab = "Measles_Cases",
356   main = "Observed_vs_Predicted_Measles_Cases_in_Nevada")
357 lines(nev_data_full$MonthDate, cases_nv, col = "red", lwd = 2)
358 legend("topright", legend = c("Observed", "Predicted"),
359   col = c("black", "red"), lty = 1, lwd = 2)
360
361 plot(cal_data_full$MonthDate, cal_data_full$CountValue, type = 'l', col =
362   "black",
363   xlab = "Year", ylab = "Measles_Cases",
364   main = "Observed_vs_Predicted_Measles_Cases_in_Cali")
365 lines(cal_data_full$MonthDate, cases_ca, col = "red", lwd = 2)
366 legend("topright", legend = c("Observed", "Predicted"),

```

```
365 | col = c("black", "red"), lty = 1, lwd = 2)
```

LISTING 1. R Code for two-state SIR model and MCMC